

AI & Machine Learning — Sample Document

What is Machine Learning?

Machine Learning is a branch of Artificial Intelligence that allows computers to learn patterns from data and make predictions or decisions without being explicitly programmed for every task.

Types of Machine Learning

- 1. Supervised Learning:** The model learns from labeled data. Common examples include Linear Regression, Logistic Regression, Decision Trees, and Random Forest.
- 2. Unsupervised Learning:** The model finds patterns in data without labeled outputs. Clustering and dimensionality reduction are common examples.
- 3. Reinforcement Learning:** An agent learns by interacting with an environment and receiving rewards or penalties.

Generative AI

Generative AI models can create new content such as text, images, audio, and code. Large Language Models use neural networks based on the Transformer architecture to understand and generate text.

RAG — Retrieval Augmented Generation

RAG combines information retrieval with a language model. Documents are converted into embeddings and stored in a vector database. When a user asks a question, relevant information is retrieved and provided to the language model as context.